# s_lolr: Model and Simulation Results

May 1, 2026

# Model

$$V^{ND}(b,l,g,s,\varepsilon) = \max_{b',l'} \left\{ u(c^{ND}) + \beta g^{1-\gamma}\, \mathbb{E}\big[\max\{V^{ND}(b',l',g',s',\varepsilon'), V^{D}(b',l',g',s',\varepsilon')\} \mid g,s\big] \right\}$$

$$c^{ND} = y(g,\varepsilon) - b - l + n(b',l',g,s) + n_l^{ND}(l',g)$$

$$V^{D}(b,l,g,s,\varepsilon) = \max_{l'} \left\{ u(c^{D}) + \beta g^{1-\gamma}\, \mathbb{E}\left[\theta V^{D}\left(\frac{b}{g},l',g',s',\varepsilon'\right) + (1-\theta)\max\left\{V^{ND}\left(\kappa\frac{b}{g},l',g',s',\varepsilon'\right),\right.\right.\right.$$

$$c^{D} = \phi_g\, y(g,\varepsilon) - l + n_l^{D}(l',g)$$

$$n_l^{ND}(l',g) = \frac{g\, l'}{R_l^{ND}}, \qquad n_l^{D}(l',g) = \frac{g\, l'}{R_l^{D}}$$

# Prices and Schedules

$$d(b,l,g,s,\varepsilon) = \mathbf{1}\left\{V^D(b,l,g,s,\varepsilon) > V^{ND}(b,l,g,s,\varepsilon)\right\}$$

$$Q(b,l,g,s,\varepsilon) = (1-d) + d\,X(b,l,g,s,\varepsilon)$$

$$X(b,l,g,s,\varepsilon) = \frac{1}{1+r^*}\,\mathbb{E}[\theta X' + (1-\theta)\left((1-e')X' + e'\kappa Q'\right)|g,s]$$

$$e'(b',l',g',s',\varepsilon') = \mathbf{1}\left\{V^{ND}(\kappa b',l',g',s',\varepsilon') \geq V^D(b',l',g',s',\varepsilon')\right\}$$

$$R(b',l',g,s) = \frac{1+r^*}{\mathbb{E}[Q' \mid g,s]}, \qquad n(b',l',g,s) = \frac{g\,b'}{R(b',l',g,s)}$$

$$p^{def}(b',l',g,s) = \mathbb{E}[d' \mid g,s]$$

$$p^{def}(b',l',g,s) \leq \bar{p}_b, \qquad n(b',l',g,s) \leq \bar{n}_g, \qquad l' \leq \bar{l}^{ND}, \bar{l}^{D}$$

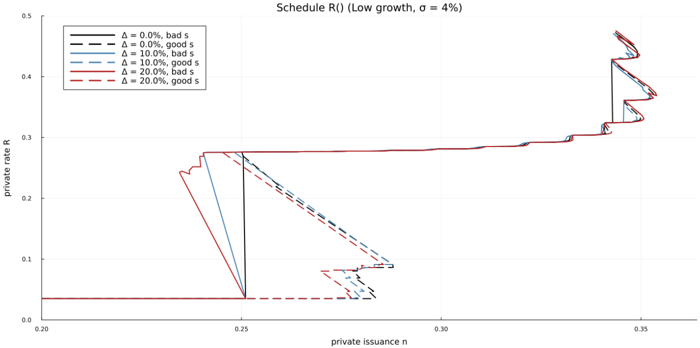
# Simulation Table

Table: Sunspot moments with and without LOLR access

| Moment | No LOLR $P_b = 1\%$ | No LOLR $P_b = 25\%$ | $\Delta = 1\%$, $P_b = 25\%$, $\sigma = 4\%$ | $\Delta = 1\%$, $P_b = 25\%$, $\sigma = 8\%$ | $\Delta = 5\%$, $P_b = 25\%$, $\sigma = 4\%$ | $\Delta = 5\%$, $P_b = 25\%$, $\sigma = 8\%$ | $\Delta = 10\%$, $P_b = 25\%$, $\sigma = 4\%$ | $\Delta = 10\%$, $P_b = 25\%$, $\sigma = 8\%$ | $\Delta = 20\%$, $P_b = 25\%$, $\sigma = 4\%$ | $\Delta = 20\%$, $P_b = 25\%$, $\sigma = 8\%$ |
|---|---|---|---|---|---|---|---|---|---|---|
| *First moments* | | | | | | | | | | |
| avg(spread) | 0.0016 | 0.0916 | 0.0909 | 0.0921 | 0.0912 | 0.0920 | 0.0925 | 0.0903 | 0.0921 | 0.0922 |
| avg(qb/y) | 0.3387 | 0.4044 | 0.4035 | 0.4051 | 0.4042 | 0.4054 | 0.4050 | 0.4031 | 0.4061 | 0.4062 |
| avg(f/y) | 0.3511 | 0.4580 | 0.4567 | 0.4589 | 0.4576 | 0.4592 | 0.4589 | 0.4560 | 0.4601 | 0.4602 |
| avg(n/y) | 0.3387 | 0.4044 | 0.4035 | 0.4051 | 0.4042 | 0.4054 | 0.4050 | 0.4031 | 0.4061 | 0.4062 |
| avg(n_l/y) | 0.0000 | 0.0000 | 0.0084 | 0.0077 | 0.0433 | 0.0408 | 0.0931 | 0.0887 | 0.1865 | 0.1791 |
| avg(l/y) | 0.0000 | 0.0000 | 0.0089 | 0.0086 | 0.0456 | 0.0448 | 0.0978 | 0.0970 | 0.1959 | 0.1953 |
| avg(b/y) | 0.3477 | 0.4131 | 0.4110 | 0.4146 | 0.4126 | 0.4144 | 0.4143 | 0.4106 | 0.4151 | 0.4149 |
| avg(tb/y) | 0.0091 | 0.0087 | 0.0080 | 0.0103 | 0.0107 | 0.0131 | 0.0140 | 0.0159 | 0.0184 | 0.0249 |
| default rate | 0.0047 | 0.1505 | 0.1536 | 0.1473 | 0.1517 | 0.1500 | 0.1496 | 0.1526 | 0.1495 | 0.1516 |
| *Low-growth state* | | | | | | | | | | |
| avg(spread) | 0.0020 | 0.0006 | 0.0014 | 0.0008 | 0.0013 | 0.0007 | 0.0008 | 0.0006 | 0.0019 | 0.0015 |
| avg(qb/y) | 0.3384 | 0.2869 | 0.2875 | 0.2871 | 0.2879 | 0.2875 | 0.2876 | 0.2875 | 0.2888 | 0.2878 |
| avg(f/y) | 0.3509 | 0.2971 | 0.2981 | 0.2975 | 0.2985 | 0.2978 | 0.2979 | 0.2978 | 0.2997 | 0.2984 |
| avg(n/y) | 0.3384 | 0.2869 | 0.2875 | 0.2871 | 0.2879 | 0.2875 | 0.2876 | 0.2875 | 0.2888 | 0.2878 |
| avg(n_l/y) | 0.0000 | 0.0000 | 0.0088 | 0.0085 | 0.0441 | 0.0425 | 0.0939 | 0.0905 | 0.1876 | 0.1809 |
| avg(l/y) | 0.0000 | 0.0000 | 0.0099 | 0.0099 | 0.0494 | 0.0493 | 0.1051 | 0.1051 | 0.2101 | 0.2102 |
| avg(b/y) | 0.3649 | 0.3100 | 0.3106 | 0.3102 | 0.3113 | 0.3104 | 0.3105 | 0.3102 | 0.3109 | 0.3102 |
| avg(tb/y) | 0.0265 | 0.0232 | 0.0242 | 0.0244 | 0.0286 | 0.0297 | 0.0342 | 0.0373 | 0.0446 | 0.0517 |
| default rate | 0.0125 | 0.3902 | 0.3880 | 0.3874 | 0.3889 | 0.3915 | 0.3944 | 0.3852 | 0.3909 | 0.3963 |
| *High-growth state* | | | | | | | | | | |
| avg(spread) | 0.0013 | 0.1108 | 0.1106 | 0.1109 | 0.1108 | 0.1109 | 0.1112 | 0.1106 | 0.1109 | 0.1108 |
| avg(qb/y) | 0.3388 | 0.4291 | 0.4291 | 0.4294 | 0.4295 | 0.4298 | 0.4289 | 0.4292 | 0.4306 | 0.4306 |
| avg(f/y) | 0.3511 | 0.4917 | 0.4917 | 0.4922 | 0.4923 | 0.4926 | 0.4917 | 0.4918 | 0.4935 | 0.4935 |
| avg(n/y) | 0.3388 | 0.4291 | 0.4291 | 0.4294 | 0.4295 | 0.4298 | 0.4289 | 0.4292 | 0.4306 | 0.4306 |
| avg(n_l/y) | 0.0000 | 0.0000 | 0.0083 | 0.0075 | 0.0431 | 0.0404 | 0.0930 | 0.0883 | 0.1863 | 0.1787 |
| avg(l/y) | 0.0000 | 0.0000 | 0.0087 | 0.0083 | 0.0448 | 0.0439 | 0.0963 | 0.0952 | 0.1929 | 0.1923 |
| avg(b/y) | 0.3373 | 0.4347 | 0.4331 | 0.4361 | 0.4347 | 0.4360 | 0.4354 | 0.4333 | 0.4368 | 0.4364 |
| avg(tb/y) | -0.0015 | 0.0057 | 0.0044 | 0.0074 | 0.0068 | 0.0096 | 0.0099 | 0.0110 | 0.0129 | 0.0194 |
| default rate | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |

Default Transition Table

# Selected Private Schedule




**Construction**

- Slice: low growth, $\sigma = 4\%$, current $b = 0$, current $l = 0$, middle $\varepsilon$.
- For each candidate private choice $b'$, search over feasible $l'$ and choose

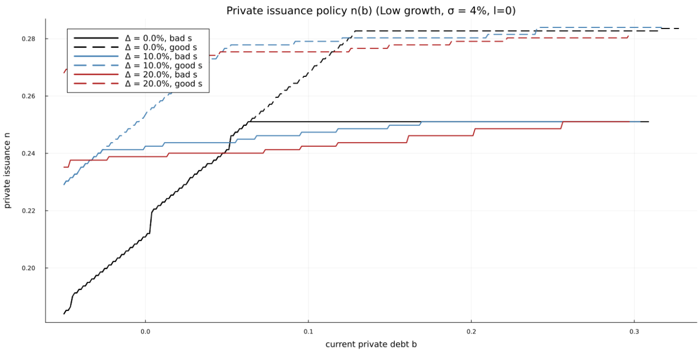
$$l^*(b') \in \arg\max_{l'} V^{ND}(b', l' \mid b = 0, l =$$

- Plot the matched pair

$$\big(n(b', l^*(b'), g, s),\ R(b', l^*(b'), g, s)\big).$$

- Colors vary $\Delta \in \{0, 10, 20\%\}$; solid is bad $s$, dashed is good $s$.

# Private Borrowing Policy




## Construction

- Slice: low growth, $\sigma = 4\%$, current $l = 0$, middle $\varepsilon$.
- For each current $b$, read the solved repayment policies

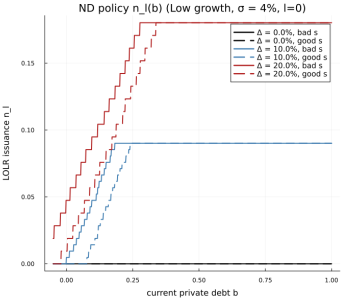
$$b^*(b),\ l^*(b)$$

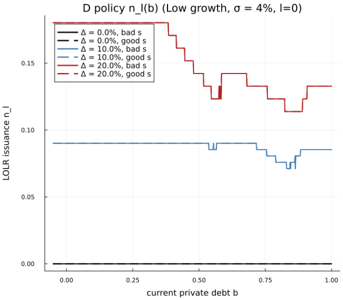
  from `b_policy_idx` and `l_policy_idx`.
- Plot

$$n(b) = n(b^*(b), l^*(b), g, s)\,.$$

- Colors vary $\Delta$; solid is bad $s$, dashed is good $s$.

# LOLR Borrowing Policy




**Construction**

- Slice: low growth, $\sigma = 4\%$, current $l = 0$, middle $\varepsilon$.
- Left panel uses repayment LOLR policy

$$n_l^{ND}(b) = \frac{g\, l^*_{ND}(b)}{R_l^{ND}}$$

from `l_policy_idx`.

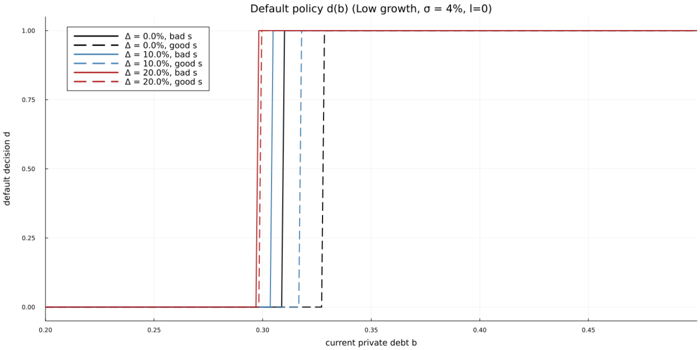
- Right panel uses default LOLR policy

$$n_l^{D}(b) = \frac{g\, l^*_{D}(b)}{R_l^{D}}$$

from `l_policy_idx_d`.

- Both panels use the full current $b$ grid.

# Default Regions




## Construction

- Slice: low growth, $\sigma = 4\%$, current $l = 0$, middle $\varepsilon$.
- Plot the solved default indicator from the Bellman comparison:

$$d(b, 0, g, s, \varepsilon) = \mathbf{1}\{V^D(b, 0, g, s, \varepsilon) > V$$

- X-axis is current private debt $b$ over the full grid.
- Colors vary $\Delta$; solid is bad $s$, dashed is good $s$.

# Appendix: Default Transition Table

Table: Default-transition shares after burn-in (percent)

| Moment | No LOLR | | $\Delta = 1\%,\ P_b = 25\%$ | | $\Delta = 5\%,\ P_b = 25\%$ | | $\Delta = 10\%,\ P_b = 25\%$ | | $\Delta = 20\%,\ P_b = 25\%$ | |
|---|---|---|---|---|---|---|---|---|---|---|
| | $P_b = 1\%$ | $P_b = 25\%$ | $\sigma = 4\%$ | $\sigma = 8\%$ | $\sigma = 4\%$ | $\sigma = 8\%$ | $\sigma = 4\%$ | $\sigma = 8\%$ | $\sigma = 4\%$ | $\sigma = 8\%$ |
| % of default starts by $g_H \to g_L$ | 61.11 | 99.93 | 99.83 | 99.86 | 99.69 | 99.89 | 99.86 | 99.83 | 99.61 | 99.72 |
| % of $g_H \to g_L$ ends in default | 1.90 | 99.79 | 100.00 | 99.89 | 99.79 | 99.86 | 99.96 | 99.79 | 99.93 | 99.79 |

Back to Main Table